

Contact

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Top Skills

Software Design
Engineering
C#

Languages

English (Native or Bilingual)
Russian (Native or Bilingual)
Hebrew (Full Professional)

Publications

Parallel Job Scheduling Using
Marginal Analysis
Workload Modeling and Internal
Backfilling for Parallel Job
Scheduling
Equipartitioning versus Marginal
Analysis for Parallel Job Scheduling

Mikhail Jacques

Software Engineer @ IAI - Israel Aerospace Industries | Developing
Real-Time Embedded Software
Center District, Israel

Summary

As a Software Engineer at IAI with over 10 years of experience, I specialize in designing and developing real-time, embedded, and desktop software solutions. My work spans the whole software engineering lifecycle, from requirements gathering and architectural design through implementation, testing, delivery, and integration, while ensuring high-quality technical documentation.

Proficient in software design, I am committed to delivering efficient and innovative solutions for UAV, robotics, and laboratory systems. I strive to contribute to high-impact projects through collaboration and technical expertise, aligning with IAI's mission and vision.

Experience

IAI - Israel Aerospace Industries

Software Engineer

December 2015 - Present (10 years 4 months)

Lod

- Specialist in developing real-time, embedded and desktop applications, with deep expertise in C and C++ programming languages, complemented by proficiency in C# and Python.
- Demonstrated ability to lead complete software engineering life cycles, including requirements gathering and analysis, architectural design, implementation, testing, delivery, integration and documentation.
- Skilled in leveraging advanced AI tools, conducting in-depth research, writing efficient and high-quality code, and producing detailed, comprehensive technical documentation.

- AI tools: OpenAI ChatGPT Plus, OpenAI Codex Programming AI, GitHub Copilot, Anthropic Claude Code, Google Gemini, Google NotebookLM, Grammarly

- Programming Languages: C, C++, C#, Python

- OS: Windows, Ubuntu, Linux Embedded, GHS Integrity RTOS

- Design Patterns: GoF design patterns in C++, C#
- Networking: Boost ASIO, TCP/UDP, RS232, RTI DDS, SOCKS, CAN bus
- Version Control: Git, MS TFS, SVN
- IDEs: Visual Studio, Eclipse, PyCharm, GHS Multi
- Source Code Editors: Visual Studio Code, Antigravity, Qt Creator
- Static Code Analysis Tools: KLOCWORK, PVS-Studio
- Management tools: Microsoft Azure DevOps, Jira
- Auxiliary tools: CMake, MS Office, Beyond Compare, Wireshark, Packet Sender, etc.

Hewlett-Packard

Software Engineer

March 2015 - December 2015 (10 months)

Yahud, Israel

Developed automated QA software tools.

Technologies: MS VS 2013, C#, WPF, StdWin, SWF, Web, Java, Eclipse, Maven, Unified Functional Testing (UFT), LeanFT, NUnit, JUnit, Mobile Android and iOS.

Development support tools: SVN, HP Agile Manager AGM.

Galil Soft Ltd

Software Engineer

March 2014 - August 2014 (6 months)

Bnei Brak, Israel

- Provided software development solutions in Linux OS environment to a variety of customers, including Intel, Verint, Giraffic.
- Embedded Linux kernel, driver and application development.
- Environment: Embedded Linux 3.0.35
- Microprocessor: phyFLEX-i.MX6 System on Module
- OS: Ubuntu 12.04 and 14.04, Fedora 20
- IDEs: Eclipse CDT, Understand Source Code Analysis & Metrics SciTool, Vim and vi editors
- Programming languages: C/C++/Python
- Tools: Oracle VM VirtualBox, Git and GitHub, Wireshark, GPSd, GStreamer, mitmproxy
- Communication protocols: HTTP, TCP/IP, NMEA

Ashdod BeYahad

Director of Operations

July 2013 - October 2013 (4 months)

Ashdod, Israel

- Planned, developed, managed and coordinated movement's field advertisement campaign.
- Was a member of the strategic and tactical planning board where planned and successfully implemented a number of strategic and tactical initiatives.
- Directly supervised work of 7 employees.
- Maintained database of potential electors of the movement and mentored others in its usage.

TrioSoft Ltd.

Embedded Software Engineer

July 2012 - July 2013 (1 year 1 month)

Ashdod, Israel

- Analyzed and consolidated customer's requirements for the development of the new Multi-Mainline Irrigation controller. The Irrigation Controller is a device used in agriculture for automatic crop management such as supplying certain dosages of irrigation water and fertilizers. The controller constitutes an integral part of a larger irrigation system that is built and marketed by Netafim Ltd.
- Implemented Irrigation Process Module from scratch in C++.

- Platform: ARM
- Real time, multithreaded environment.
- Environment: Synchronous Multi-threaded environment
- Architectural pattern: Asynchronous, non-blocking, one-way messaging
- IDEs: MS Visual Studio 2012 and IAR Embedded Workbench
- Programming languages: C/C++/C#
- API: CMX RTOS, EASTL
- Microprocessor: LPC2212/2214 single-chip 16/32-bit ARM micro-controller

Freelancer

2 years 4 months

English Language Teacher (part-time)

September 2011 - June 2013 (1 year 10 months)

Ashdod

- Taught English language on a part-time basis to elementary and high-school students and to adult professionals.
- Assisted graduate students with writing their dissertation papers.

Analyst and Web Developer (part-time)

March 2011 - March 2012 (1 year 1 month)

Ashdod

- Analyzed requirements, developed content, built, tested and hosted websites for various customers.
- Technologies: MS Visual Studio 2010, Adobe Dreamweaver CS6, HTML, PHP, phpMyAdmin, MySQL, JavaScript, CSS, Google Analytics,

Optoacoustics Ltd

Project Manager

April 2011 - June 2011 (3 months)

Moshav Mazor, Israel

- Oversaw the construction of the company's new site at Moshav Mazor and coordinated the company's transition from Or Yehuda site to Moshav Mazor site.

Gimmonix

Web Developer

February 2011 - April 2011 (3 months)

Tel-Aviv, Israel

- Programmed server-side business logic for rental cars online comparison and booking engine for Online Travel Agencies.
- Technologies: MS Visual Studio 2010, .NET framework, C#, XML, HTML, JavaScript.

Trent University

Lecturer in Computer Science

April 2010 - August 2010 (5 months)

- Developed course curriculum and presented course content on Management Information Systems.
- Developed student labs in Verilog HDL.
- Judged youth science projects at the Canada-Wide Science Fair (CWSF).

Ontario Provincial Police

Provincial Constable

January 2009 - December 2009 (1 year)

- Investigated motor vehicle collisions, incidents of domestic violence, assaults, impaired operations of motor vehicles, break and enters, thefts, mischief to property, prank phone calls, suspected frauds and a large scale multijurisdictional credit card fraud case.
- Apprehended, arrested, investigated and charged offenders, referred individuals needing help to community and other service agencies.

- Documented events and reported to superiors on emergencies, disruptions, and violations.
- Technologies: Niche RMS, CPIC.

Trent University

Lecturer in Computer Science

January 2007 - August 2008 (1 year 8 months)

- Developed and taught Software Engineering courses (C, C++, C# programming languages) to classes of up to 100 students.
- Developed and taught digital circuitry logic and fundamentals of computer design: MIPS architecture, Verilog HDL.
- Developed course curriculum and presented course content on Management Information Systems and Information Systems Analysis and Design courses with a focus on successful implementation for government and businesses.
- Designed student labs and ensured Teaching Assistance's compliance to schedule, quality and content.
- Successfully marketed Online Academic Support and Teaching Tools for faculty and students.

Trent University

Computer Science Teaching Assistant

September 2004 - April 2006 (1 year 8 months)

- Provided teaching assistance and conducted lectures and seminars on behalf of Computer Science department faculty members.

Trent University

2 years 9 months

Project Manager and Software Engineer

August 2003 - May 2004 (10 months)

Ontario, Canada

- Pioneered, managed, designed, implemented, tested and delivered web-based Booking and Information System for University/Community Recreational & Athletics Complex.
- Led and managed a team of 3 software developers.
- Technologies: MS Project, Adobe Dreamweaver, HTML, CSS, JavaScript, PHP, MySQL, phpMyAdmin, Apache.

Computer Science Research Assistant

May 2002 - May 2004 (2 years 1 month)

Ontario, Canada

- Developed and implemented computer-based simulation models to investigate the performance of parallel job scheduling strategies.

Computer Science Lab Advisor

September 2001 - April 2004 (2 years 8 months)

- Provided pedagogical help for 1st and 2nd year University students with respect to their computer science and math school work.

PlanPlus Inc.

Analyst and Software Tester

April 2003 - August 2003 (5 months)

Ontario, Canada

- Collected and analyzed economic data.
- Tested software of a large accounting and financial system delivered in a web-based environment.

Intel Corporation

Project Coordinator

July 1997 - September 2000 (3 years 3 months)

USA (New Mexico, Oregon), Israel

- Oversaw and coordinated proper design of systems for electronic tool-sets.
- Documented requirements and responded to changing business requirements.
- Conducted various supervisory tasks to ensure problem-free installation of electronic, electrical, tele-communication and mechanical toolsets and sub-systems.
- Programmed for MS Excel in Visual BASIC for Applications.
- Delivered on time and on budget.

ZIM Intergrated Shipping Service LTD

Electrical Officer

May 1996 - July 1997 (1 year 3 months)

- Maintained secure operational condition of electrical systems.
- Responded to malfunctions of and troubleshoot electrical, refrigeration, naval control and navigation systems.
- Developed cultural sensitivity expertise when serving on merchant cargo ships sailing between Europe and the Far East.

Israeli Navy

Marine Electrician

May 1993 - April 1996 (3 years)

Israel

- Maintained secure operational condition of electrical systems.
- Responded to malfunctions of and troubleshoot naval electrical systems.
- Responded to emergencies by providing front-line electrical circuitry repair service to special marine forces.

Education

Ontario Police College, Aylmer, Ontario, Canada

Diploma, Basic Constable Training · (2008 - 2008)

Trent University, Peterborough, Ontario, Canada

Master of Science, Applications of Modeling in the Natural & Social Sciences · (2004 - 2006)

Trent University, Peterborough, Ontario, Canada

Bachelor of Science, Computer Science · (2001 - 2004)

Israel Ministry of Transport

Certificate of Marine Electrical Officer Second Grade, Marine Electricity · (1996 - 1997)

"Mevoot Yam" Marine Technological College, Mikhmoret, Israel

Diploma in Marine Electricity and Control, Marine Electricity and Control · (1992 - 1993)